

TEXT GENERATION

```
# Uninstall the current Transformers library
!pip uninstall -y transformers

# Install the latest stable version
!pip install -U transformers
```

```
# transformer = Hugging face python library that provides the transformer based models

#
===IMPORTING===

# AutoModelForCausalLM = Automatically selects and loads the suitable causal LM
# AutoTokenizer = Automatically loads the suitable tokenizer for the loaded model
# pipeline = creates an interface for using the model
# torch = transformers uses the PyTorch as the backend for running the models
from transformers import AutoModelForCausalLM , AutoTokenizer , pipeline
import torch

# Model name on Hugging face you want to load
model_name="microsoft/Phi-3-mini-4k-instruct"

model=AutoModelForCausalLM.from_pretrained( # Automatically loads the pretrained model
    model_name,
    device_map="cuda", # Loads the model on gpu
    torch_dtype="auto", # Automatically selects the datatype for computation
    trust_remote_code=True, # Allows the custom code to run if needed
)

# Load the tokenizer that automatically converts the text into tokens and generated tokens into back to text
tokenizer=AutoTokenizer.from_pretrained(model_name)

generator=pipeline(
    "text-generation", # Tells the transformers to generate the text
    model=model,
    tokenizer=tokenizer, # Previously loaded tokenizer and model
    return_full_text=False, # Output contains only newly generated tokens not the input ones
    max_new_tokens=500, # MAX no.of. new tokens to generate
    do_sample=False # This tells how the next token is generated (Always chooses the highest probability)
)

messages=[
    { "role":"user" , "content" : "Create a funny joke about chickens."} # Chat style input It has 2 keys "role" = Who
]

output=generator(messages) # Pipeline internally does Converts chat messages into the model's expected prompt
print(output[0]["generated_text"]) # Model generates a list of output select the first generated output
```

The cache for model files in Transformers v4.22.0 has been updated. Migrating your old cache. This is a one-time only operation.
0/0 [00:00<?, ?it/s]

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:

The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as an environment variable, or add it to your Colab secrets. You will be able to reuse this secret in all of your notebooks.

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    model_name,
    device_map="cuda",
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)
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generator=pipeline(
    "text-generation",
    model=model,
    tokenizer=tokenizer,
    return_full_text=False,
    max_new_tokens=500,
    do_sample=False
```

```
)

messages=[
    { "role":"user" , "content" : "What is AI."}
]
```

```
output=generator(messages)
print(output[0]["generated_text"])
```

WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b323eae81227eaba4464caced4e.modeling_phi3:`flash-attn` is not installed. Using the default attention implementation, which may be slower. You can install `flash-attn` to use the flash attention implementation.

WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b323eae81227eaba4464caced4e.modeling_phi3:Current device (cuda:0) is not available. Falling back to CPU. This may significantly slow down training.

Loading checkpoint shards: 100% 2/2 [00:30<00:00, 14.30s/it]

AI, or artificial intelligence, refers to the development of computer systems that can perform tasks that typically require human intelligence.